OFFICEPULSE CHURN-PREVENTION CRM — BUILT ON DATABRICKS

See churn 90 days before the customer says goodbye

For the executive sponsor (CEO / CFO) and the domain owner (VP Customer Success)

Working prototype · synthetic data · fictional names (CloudNest / OfficePulse)

THE OUTCOME

Executive summary

+¥44.5M
ARR retained / year

+2.1 pts
GRR uplift

60.6%
of churning ARR flagged at T-90 days (today: 0%)

- **+¥44.5M ARR retained per year** (≈ $300k) — save plays start 90 days before renewal instead of after the cancellation e-mail
- **Gross Revenue Retention 84.6%** → **86.7%** (+2.1 pts) on a ¥2.19B ARR base
- **36 high-risk accounts (¥102M ARR) are not on any CSM's radar today** — the cockpit surfaces them with the reasons and a playbook
- **Ask:** approve a 90-day pilot with two CSM pods (A/B on flagged accounts) to confirm the save rate before scale-out

WHY CHURN IS A BOARD-LEVEL ISSUE

The problem, in your KPIs

¥330M
ARR churned (12 m)

84.6%
Gross Revenue Retention

31 days
median warning before renewal today

- **¥330M ARR and 93 logos lost in the last 12 months** — GRR 84.6%
- Only **47%** of churners ever tell us; the cancellation ticket arrives a median **31 days** before renewal — too late for a save play
- The warning signs exist but live in three silos: product usage (GA4), support (Zendesk), contracts (CRM)
- The current rule-based health score is hand-tuned, not explainable per account, and misses half of the churners

RISK YOU CAN ACT ON NOW

What the model sees in the book today

41
high-risk accounts

36
not flagged by CSMs

¥280M
expected ARR at risk

- **41** accounts with ≥ 50% probability of not renewing — **¥120M** ARR; **¥96M** of it renews within 180 days
- **36 of the 41** are still marked *active* by their CSM — **¥102M** ARR at risk is invisible today
- Expected ARR at risk (Σ probability × ARR) = **¥280M** — a number Finance can forecast against
- Example — Asuka Construction (Enterprise, ¥13.3M ARR, renews 25 Dec): CRM status *active*, model 67%. Drivers: sensor detection rate 61%, CSAT 3.0, logins 57%

MODEL RESULTS, VALIDATED OUT OF TIME

Proof: earlier and more precise than today

Out-of-time test (renewals after 1 May 2026)	Status quo (customer's ticket)	Old rule score	ML model
Churning ARR flagged 90 days before renewal	0%	53.9%	**60.6%**
Alert precision at T-90 (same alert volume)	—	59%	**63%**
Precision in top-10% riskiest snapshots	—	40%	**76%**
ROC AUC / PR AUC	—	0.772 / 0.393	**0.847 / 0.621**

Trained only on renewals whose outcome was known before 1 May 2026, tested on the 247 renewals after it (47 churned). Rules compared at the same number of alerts. Full outputs: evidence/notebook_runs/train_register_serve.ipynb

VALUE FOR THE DOMAIN OWNER

For the VP Customer Success: a different Monday

- **Monday morning:** the CS cockpit lists renewals by probability, with a *missed-risk* filter (high risk, not flagged)
- **Why:** top-3 drivers per account (SHAP) — e.g. sensor detection rate, CSAT, urgent tickets, auto-renew off
- **What if:** move the levers (re-provision sensors → detection 85%, resolve tickets) and re-score live on Model Serving
- **What to do:** a generated playbook — prioritized actions, owner, due date, talk track, success metric — one click to save as an action or meeting
- **Workload:** ≈ 20 T-90 alerts per quarter at 63% precision — a short, explained list instead of the whole book
- **Ask anything:** Genie answers ad-hoc questions in Japanese over the same governed data (e.g. ‘high-risk accounts renewing in 90 days by industry’)

VALUE FOR THE EXECUTIVE SPONSOR

For the CEO / CFO: retention you can forecast

- **Retention:** +¥44.5M ARR per year retained, GRR +2.1 pts — compounding into NRR and valuation
- **Predictability:** expected ARR at risk (¥280M) is refreshed weekly per account — renewal forecast stops being a surprise
- **Root causes for Product:** detection-rate drops (sensor health) are the #1 churn driver — a fixable product/ops issue, not just a CS issue
- **Leverage:** same CS headcount covers the at-risk book 90 days earlier; no new tooling licences — one governed platform
- **Low risk to start:** runs on serverless (scale-to-zero serving, autoscaling Lakebase); pilot uses existing GA4 / Zendesk / CRM exports

CONSERVATIVE, TRANSPARENT, ADJUSTABLE

Business case and assumptions

Save rate when a play starts ≥ 90 days before renewal	ARR retained / year	Increment vs today	GRR
20%	¥40.0M	+¥24.5M	85.8%
30% (base case)	¥60.0M	**+¥44.5M**	**86.7%**
40%	¥80.0M	+¥64.5M	87.6%

- Base: trailing-12-month churned ARR ¥330M on a ¥2.19B starting ARR (Lakeflow gold tables)
- Early coverage = 60.6% of churning ARR flagged at T-90 (out-of-time test)
- Today: plays start only after a cancellation ticket (47% of churners), save rate 10% → ¥15.4M/year
- Assumptions are parameters of the business-impact notebook — the pilot replaces them with measured save rates

ARCHITECTURE

How it works: one integrated data journey



Unity Catalog — one governed namespace: tags · certified gold · PII masks · grants · lineage · model registry · AI Gateway model service

- One Lakeflow Job runs the whole journey (generate → pipeline → govern → train/register/serve → score → publish → business case)
- Gen AI is grounded on the model's probability and drivers, called through a Unity Catalog model service (governed, permissioned)
- Genie space over certified gold tables, with instructions, sample questions and SQL benchmarks; embedded in the app

GOVERNANCE & SECURITY

Trust: governed from raw file to recommendation

- **Unity Catalog**: raw volume, 12 pipeline tables, ML tables, model, LLM service and mask function in one schema; gold tables tagged *certified*
- **PII**: contact e-mail/phone classified (class.* tags) and masked by a UC column mask declared in the pipeline
- **Least privilege**: the app's service principal gets read on the schema, EXECUTE on the model service, CAN_QUERY on the endpoint, its own Postgres role
- **Lineage & audit**: raw → bronze → silver → gold → scores captured automatically; MLflow run + model version on every score
- **Data quality**: 130 defective GA4 rows (duplicates, missing tenant, impossible counts) dropped by expectations and reported in the event log
- **Responsible AI**: the model scores, the LLM explains; every recommendation cites the drivers it is based on

INSIGHTS FOR THE WHOLE COMPANY

What drives churn — and who can fix it

Driver (feature importance rank)	What it means	Owner of the fix
1. Sensor detection rate (4 weeks)	Devices offline / mis-configured → product feels broken	Support + Field Ops
2. Average CSAT (12 weeks)	Poor support experience precedes churn	Support leadership
3. Auto-renew switched off	Procurement already reviewing the contract	Account Executive
4. Urgent tickets (12 weeks)	Escalations accumulate before a decision	Support
5. Login trend (8 weeks)	Adoption fading after rollout	CSM (enablement)

FROM PROTOTYPE TO MEASURED VALUE

Next steps: a 90-day pilot

- **Weeks 0–2**: connect live GA4, Zendesk and CRM via Lakeflow Connect; switch publish to Lakebase Synced Tables
- **Weeks 2–10**: pilot with two CSM pods — half of flagged renewals get the playbook (A/B); log every action in the cockpit
- **Week 12**: read-out on measured save rate, GRR and CSM time; retrain monthly with the new outcomes
- **Success criteria**: ≥ 25% save rate on early-flagged ARR; ≥ 55% of churning ARR flagged at T-90; CSM adoption ≥ 80% weekly
- **Decision needed today**: executive sponsor + VP CS nominate pilot pods and agree success criteria

FOR THE TECHNICAL REVIEWER

Appendix: model and data details

- Label: account does not renew at its next annual renewal; scored weekly; evaluated 30–180 days before renewal
- Features (20, no look-ahead): usage rates and trends, sessions, tickets/urgent/technical/cancellation, slow responses, CSAT, open tickets, auto-renew, plan
- Excluded on purpose: seats/ARR (identity leakage), days-to-renewal/tenure (not actionable); accounts with < 12 weeks of history are not scored
- Model: GradientBoosting (depth 2, 250 trees) wrapped as an MLflow pyfunc returning probability + SHAP top-3 drivers
- Data: 612 synthetic accounts, 78 weeks of GA4, 4.4k Zendesk tickets with 15k comments, 18 months of CRM contracts
- Evidence (text): notebooks with outputs, pipeline event log, Genie Q&A, live app API calls — see repository /evidence